

The empirical recurrence for OEIS A297651: recovering a conjecture that is not written down

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Abstract

OEIS A297651 records “Empirical recurrence of order 60”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 89$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 60, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 60 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297651 is “Number of $n \times 5$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 1, 2 or 3 neighboring 1s.”. It has offset 1 and begins

12, 494, 6838, 131044, 2525244, 46187510, 864944851, 16149058068, ...

The entry states:

Empirical recurrence of order 60 (see link above)

As of the “Last modified” line on the live entry (Jan 02 2018 15:12:20, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 89$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 89$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 178$ exact terms of the model returns order 60 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{60} c_i a(n-i),$$

c_1	9	c_{16}	4452657263	c_{31}	305771363891	c_{46}	459115977710
c_2	81	c_{17}	−12479256576	c_{32}	−159861612537	c_{47}	−143490913736
c_3	1580	c_{18}	−11889306832	c_{33}	81719847248	c_{48}	−106116482342
c_4	4866	c_{19}	2978190646	c_{34}	−1213924502193	c_{49}	69936611944
c_5	18956	c_{20}	59732601301	c_{35}	−68992576629	c_{50}	−15752272132
c_6	−254956	c_{21}	−47517250218	c_{36}	−1121024546538	c_{51}	16391636976
c_7	−1161211	c_{22}	80312290563	c_{37}	389287385224	c_{52}	−1998671456
c_8	−1032855	c_{23}	−254879995736	c_{38}	1621540572107	c_{53}	−1692451688
c_9	18415367	c_{24}	5623258877	c_{39}	663589362720	c_{54}	2296335360
c_{10}	58555200	c_{25}	−440591953881	c_{40}	396180429536	c_{55}	254319392
c_{11}	−25423522	c_{26}	608801591239	c_{41}	−523909066460	c_{56}	−8218432
c_{12}	−564504402	c_{27}	179718964261	c_{42}	178405285519	c_{57}	−58431360
c_{13}	−1086399261	c_{28}	532178577951	c_{43}	117862754277	c_{58}	−1048064
c_{14}	1253795432	c_{29}	−426695243194	c_{44}	−615190505128	c_{59}	−1445376
c_{15}	7515584072	c_{30}	−81325341043	c_{45}	−475101195581	c_{60}	1296384

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 60, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $60 + 4$ terms removed returns order 60 as well, so $\deg q_{\min} = 60 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $60 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 60 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A297651.
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